

**Problem 8.17.** Let  $A$  be the language of properly nested parentheses. For example,  $(( ))$  and  $((()()))()$  are in  $A$ , but  $)()$  is not. Show that  $A$  is in  $L$ .

*Proof.* To show that  $A \in L$ , we give a log space algorithm  $M$  that decides  $A$ .

1. Let  $c$  be a binary integer initialized with 0.
2. Repeat for each input symbol  $b$  from left to right.
3.     If  $b$  is  $($ , then increment  $c$  by 1.
4.     If  $b$  is  $)$  and  $c > 0$ , then increment  $c$  by 1.
5.     If  $b$  is  $)$  and  $c = 0$ , then *reject*.
6. If  $c = 0$ , then *accept*. Otherwise, *reject*."

□